

KAREN XIAO

Curriculum Vitae

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EDUCATION

Wellesley College, *Wellesley, MA* 2023 - 2026
B.A. in Computer Science and Mathematics, *Summa Cum Laude*
Departmental Honors in Computer Science
Thesis Advisor: Brian Brubach

University of Maryland, College Park, *College Park, MD* 2022 - 2023
Honors College, Gemstone Research Program

RESEARCH EXPERIENCE

Undergraduate Honors Thesis, *Wellesley College* Sep 2025 - May 2026
Thesis Title: Quantifying Diversity of Educational Evaluation Using Relative Student Performance
Advised by Professor Brian Brubach

NSF REU Intern, *CMU Human-Computer Interaction Institute* May 2025 - Aug 2025
Project: Exploring the Impact of Data-Driven Optimization of Tutoring Systems on Student Learning
Advised by Conrad Borchers and Professor Vincent Aleven

Research Assistant, *Institute for Mathematics and Democracy* Dec 2024 - Sep 2025
Project: Empirical Investigation of Ranked Voting Methods
Advised by Professors David McCune, Ismar Volić, and Andy Schultz

Undergraduate Student Researcher, *Wellesley College CrowdNets Lab* Aug 2024 - May 2025
Project: A Graph-Based Spatio-Temporal POI Ranking Measure for Pickup and Delivery Platforms
Advised by Professor Christine Bassem

Undergraduate Student Researcher, *MIT Urban Metabolism Group* Feb 2024 - May 2024
Project: Urban Climate Policy Mapping with AI
Advised by Norhan Bayomi

PUBLICATIONS

- Liu, Q., Borchers, C., Xia, M., **Xiao, K.**, Carvalho, P., Koedinger, K., & Aleven, V. (2026). Evaluating a Process for the Datadriven Redesign of Educational Technology: Improved Learning Processes. In *Proceedings of the 27th International Conference on Artificial Intelligence in Education (AIED '26)*. <https://doi.org/10.1007/978-3-032-29760-0.12>
- Xiao, K.**, Irisumi, M., & Bassem C. (2025). A Graph-Based Spatio-Temporal POI Ranking Measure for Pickup and Delivery Platforms. In *Proceedings of the 33rd ACM International Conference on Advances in Geographic Information Systems (SIGSPATIAL '25)*. <https://doi.org/10.1145/3748636.3766535> [**SRC 2nd Place Winner**]

TEACHING EXPERIENCE

Teaching Assistant

Wellesley College
CS 231 Design & Analysis of Algorithms Spring 2024 - Spring 2026
STAT 218 Introductory Statistics & Data Analysis Spring 2026

HONORS AND AWARDS

Awards

Excellence in Teaching Award in Computer Science, *Wellesley College* May 2026
NSF GRFP Honorable Mention, *National Science Foundation* Apr 2026

Undergraduate SRC 2nd Place Winner, *ACM SIGSPATIAL '26*

Nov 2025

Conference Funding

SIGSPATIAL Travel Award (SIGSPATIAL '26) - \$1,000, *National Science Foundation*

2025

Science Center Student Conference Travel Grant - \$300, *Wellesley College*

2025

WORK EXPERIENCE

Financial Data Analyst Intern, *Quantbot Technologies*

Summer 2023, 2024, 2026

Team: Data Trading Lab

LEADERSHIP EXPERIENCE

Co-Captain, *Wellesley Club Squash Team*

2025-2026

Networking Chair, *Wellesley Computer Science Club*

2024-2026